

DWARAKANATH C

GCP Cloud Developer & AI Engineer | Senior Backend Engineer (Golang / Java)
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PROFESSIONAL SUMMARY

Senior Backend Software Engineer and GCP Cloud Developer with 10+ years of experience across backend development, event-driven microservices, and cloud/AI engineering, including 4+ years of hands-on Golang and 6+ years in Java/J2EE. Builds high-throughput distributed systems with Go, Kafka, RabbitMQ, PostgreSQL, Redis, and Docker/Kubernetes for regulated platforms in financial services, travel, and FMCG. Currently employed by Cognizant (from 08/2024) and designs production AI platforms on GCP for Rentokil Initial using Vertex AI, Gemini, Agent Engine, Document AI, RAG, BigQuery Conversational Analytics, Cloud Run, Pub/Sub, and Terraform across document intelligence, sales prioritisation, cash allocation, and natural-language analytics over portfolio, financial, and service-line data. Previously delivered Golang loyalty services for American Express on the same Cognizant engagement. Comfortable working across time zones and contributing to architecture, code quality, observability, and delivery.

TECHNICAL SKILLS

- Languages: Golang, Java (Spring Boot, JPA), Python, SQL
- Backend & APIs: RESTful services (net/http, Gorilla Mux, Gin, Echo, Spring Boot)
- Event-Driven Systems: RabbitMQ, Kafka, Pub/Sub (message queues, event processing, async workflows)
- Databases & Cache: PostgreSQL, Redis, MongoDB, MySQL, Oracle, BigQuery
- Containers & Orchestration: Docker, Kubernetes (deployment, health checks, resource limits, autoscaling)
- Cloud & DevOps: Google Cloud Platform (Cloud Run, BigQuery, Pub/Sub, IAM, Terraform, API Gateway, Cloud Scheduler, Secret Manager), AWS, OpenShift Hydra, Jenkins, Spinnaker
- AI & Data: Agentic AI, RAG, Vertex AI, Agent Engine, Gemini, Google ADK, Document AI, Conversational Analytics, machine learning, LLM inference (vLLM, Gemma), BigQuery SQL, QlikCloud
- Quality & Observability: Unit testing, BDD (Cucumber), Zap Logger, Splunk, Elastic, Kibana
- Practices: Microservices, Event-Driven Architecture, Clean Architecture, Agile/Scrum, Code Reviews

EDUCATION

Master's in Computer Application (M.C.A) - SVU, India - 2012

Bachelor's in Computer Science (BSc) - SVU, India - 2009

PROFESSIONAL EXPERIENCE

Cognizant, UK | 08/2024 to current

Project 1: Cloud & AI Platform Engineering - Client: Rentokil Initial, UK (current) | Team size: 8 | 03/2026 to current

Role: GCP Cloud Developer & AI Engineer

- **Consumables - Document Intelligence:** Designed an event-driven platform that converts incoming safety and chemical PDFs into structured analytics using Drive webhooks, API Gateway, Cloud Run, Pub/Sub, Document AI, BigQuery, Cloud Storage, and QlikCloud.
- **Consumables - Reliability and RAG:** Separated fast webhook ingestion from long-running AI workers, added RAG over extracted BigQuery entities and layered dead-letter recovery across Cloud Storage, BigQuery error rows, and Pub/Sub retry topics.
- **EU Prospect Prioritisation:** Delivered a three-CTE BigQuery model covering employee, occupation, and prospect data so ranking includes only leads assigned through employee lead codes.
- **Multi-agent Sales Intelligence:** Built Vertex AI Agent Engine services for data retrieval and RAG enrichment, ML-based value estimation, and prospect prioritisation, orchestrated with Gemini and Google ADK.
- **AI Cash Allocation:** Built an event-driven credit-control workflow that classifies inbound Gmail messages with Gemini on Vertex AI and writes structured results to a BigQuery audit trail.

- **AI Cash Allocation - Agentic Matching:** Designed a RAG workflow over invoice history to match remittances, propose cash allocations, and route processing through Pub/Sub and Cloud Run.
- **iCABS Portfolio Agent:** Published a BigQuery Conversational Analytics agent over monthly iCABS portfolio snapshots so Group Strategy can ask where new sales and live value land by sector and country, without a manual extract.
- **Cognos Financial Performance Agent:** Landed mapped Cognos P&L in BigQuery and built natural-language Q&A for Europe revenue, profit, and margin by country and business line, replacing recurring Excel pivots.
- **Service Line Penetration Agent:** Built a Europe cross-sell agent that classifies customers as pest only, hygiene only, or both, by country and sector, from monthly service-line snapshots rather than campaign rates.
- **LLM Inference PoC:** Led a self-hosted Gemma deployment on Cloud Run GPU with vLLM to evaluate a cost-efficient alternative to third-party inference APIs.
- **Platform Quality:** Improved reliability through Terraform-managed infrastructure, autoscaling, data-quality reviews, alerting, code reviews, and documented recovery paths.

Tech: GCP (Vertex AI, Agent Engine, Gemini, Document AI, BigQuery, Cloud Run, Pub/Sub, Cloud Storage, IAM, API Gateway, Cloud Scheduler, Secret Manager), RAG, Google ADK, Terraform, Python, SQL, QlikCloud, vLLM, Gemma

Project 2: Global Loyalty Program - Client: American Express, UK | Team size: 7 | 08/2024 to 04/2026

Role: Senior Golang Developer

- Design and develop scalable Golang microservices supporting loyalty enrolments and reward processing platforms, handling millions of transactions daily.
- Build RESTful APIs using net/http and Gorilla Mux, focusing on clean design and maintainability with comprehensive unit test coverage.
- Implement event-driven workflows using Kafka to process lifecycle, eligibility, and reward events with guaranteed delivery mechanisms.
- Optimise services for high throughput (10,000+ requests/minute), low latency, and resilience in distributed systems.
- Work with PostgreSQL, MongoDB, and Redis for transactional data handling, caching strategies, and performance optimization.
- Deploy and manage containerised services using Docker and Kubernetes, including configuration of health checks, resource limits, and auto-scaling policies.
- Lead architecture discussions for microservice decomposition, API design patterns, and event-driven messaging strategies across cross-functional teams.
- Write comprehensive unit tests and BDD test suites, perform code reviews, and support production incident resolution with root cause analysis.

Tech: Golang, Kafka, PostgreSQL, Redis, Docker, Kubernetes, Microservices, Git, Elastic, Kibana, OpenShift Hydra

Project 3: Campaign Scheduler - Expedia, UK | Team size: 6 | 05/2022 to 07/2024

Role: Golang Developer

- Developed backend services in Golang for campaign scheduling and customer engagement platforms processing 500,000+ scheduled events daily.
- Implemented asynchronous messaging using RabbitMQ for job queues, event processing, and inter-service communication, handling 100,000+ messages/day with guaranteed delivery and retry mechanisms.
- Designed and integrated event-driven workflows for communication and calendar scheduling services using RabbitMQ exchanges and routing patterns.
- Implemented concurrent and asynchronous processing using goroutines and channels to handle high-volume scheduling workloads efficiently.
- Designed and developed RESTful APIs using Echo and net/http with comprehensive error handling and input validation.
- Built timezone-aware scheduling logic for global users supporting 50+ timezones with daylight saving adjustments.
- Deployed containerised services using Docker on AWS infrastructure with CI/CD pipelines via Jenkins.

- Performed code reviews and contributed to improving service stability and performance through observability implementations.

Tech: *Golang, RabbitMQ, Microservices, Echo, net/http, AWS, Docker, MongoDB, Jenkins*

Project 4: Relocation Project - Aires Mobility, US | Team size: 6 | 09/2021 to 05/2022

Role: Golang Developer

- Developed backend APIs for a Benefits Management System using Golang with focus on clean architecture principles.
- Designed microservices architecture, API contracts, and data models from scratch following RESTful best practices.
- Authored comprehensive technical documentation and implemented unit and integration tests achieving 85%+ code coverage.
- Participated in sprint planning, daily standups, and peer code reviews in an Agile environment.

Tech: *Golang, Microservices, AWS, Docker, PostgreSQL*

Project 5: iGTB Payment Service Hub (IPSH) - Santander, Spain | Team size: 10 | 03/2019 to 08/2021

Role: Software Developer

- Developed RESTful backend services using Java and Spring Boot for enterprise banking and payment systems in a highly regulated financial environment.
- Implemented business validation rules using Drools Engine ensuring compliance with financial regulations and industry standards.
- Worked on country-specific data mappings and regulatory compliance features including SEPA, SWIFT, and local payment schemes.
- Followed Agile and TDD practices, collaborating closely with QA, compliance, and business teams to ensure quality delivery.

Tech: *Java 8, Spring Boot, JPA, Microservices, Docker, AWS, Swagger, Git, PostgreSQL*

PERSONAL PROJECTS

- **LifeHub (In progress):** Built an end-to-end personal productivity platform with FastAPI, PostgreSQL, Gemini priority classification, Next.js, and Expo; the Todos workflow is complete, with Calendar, meeting notes, and Google OAuth planned next.
- **Maths App (Shipped v1):** Developed a Grade 1 Android learning app in Kotlin and Jetpack Compose covering place value, expanded form, ordinals, addition, and subtraction, with Learn/Practice modes and instant scoring feedback.
- **The AI Pulse (Shipping daily):** Built a daily AI-engineering brief pipeline in Python using Groq, OpenRouter, LLM relevance scoring, and Gmail delivery to produce concise builder- and leadership-focused summaries.

Tech: *Python, FastAPI, PostgreSQL, Gemini, Next.js, TypeScript, Expo, Kotlin, Jetpack Compose, Groq, OpenRouter, Gmail*

AWARDS AND CERTIFICATIONS

- Best Performer of the Month Award from SmartLoyal
- AWS Certified Developer Associate - Dec 2023
- GCP Professional Cloud Architect (in progress)
- Spring Boot and Microservices with Spring Cloud (Udemy)
- GEM Award from Polaris for Excellent Performance
- Spot Excellence Award from Polaris
- Agile Software Development (Coursera)

EARLIER PROJECTS

- Interac-e-Transfer (Mobile Application) as Developer: 07/2017 – 02/2019
- Branch Profitable Index (BPI) as Developer: 09/2015 to 06/2017